VANISHING OF HOM FROM A PERFECT OBJECT AGAINST THE TRUNCATION TRIANGLE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7260-7273, immediately following the statement at 7252-7259.

Editorial note. The displayed Hom identity is the reduction step; the vanishing input is cited once.

Allowed standard prerequisites (editorial).

- Ext vanishing from perfect into bounded below. Lemma lemma-ext-from-perfect-into-bounded-QCoh of this chapter, cited at proof lines 7270-7272. Terminal input: Hom(P, F[i]) = 0 for i large when P is perfect and F bounded below in D_QCoh.

Source reference. Original lemma 09MB: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4328–4328. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $P \in D_{perf}(\mathcal{O}_X)$ and $E \in D_{QCoh}(\mathcal{O}_X)$. Let $a \in \mathbf{Z}$. The following are equivalent*

(1) $\mathrm{Hom}_{D(\mathcal{O}_X)}(P[-i], E) = 0$ *for* $i \gg 0$, *and*
(2) $\mathrm{Hom}_{D(\mathcal{O}_X)}(P[-i], \tau_{\geq a}E) = 0$ *for* $i \gg 0$.

Proof. Using the triangle $\tau_{<a}E \to E \to \tau_{\geq a}E \to$ we see that the equivalence follows if we can show

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(P[-i], \tau_{<a}E) = \mathrm{Hom}_{D(\mathcal{O}_X)}(P, (\tau_{<a}E)[i]) = 0$$

for $i \gg 0$. As P is perfect this is true by Lemma 09MB. □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.